

ADA BOOST

ALG: $\alpha_i = \frac{1}{N}$

$t = 1$ to T :

← iteracije
na modelu
u ANS.

LEARN $f_t(x; \alpha^t)$ ← WEAK LEARNER

$$WE_t = \sum_i \alpha_i \cdot \mathbb{I}(\hat{y}_{t,i} \neq y_i)$$

← ACCURACY

$$W_t = \frac{1}{2} \ln\left(\frac{1 - WE_t}{WE_t}\right)$$

$$\alpha_i^{t+1} = \begin{cases} \alpha_i^t \cdot e^{-W_t}, & f_t(x_i) = y_i \\ \alpha_i^t \cdot e^{W_t}, & f_t(x_i) \neq y_i \end{cases}$$

← FACTOR

$$\alpha_i^{t+1} = \alpha_i^t \cdot e^{-W_t \cdot f_t(x_i) \cdot y_i}$$

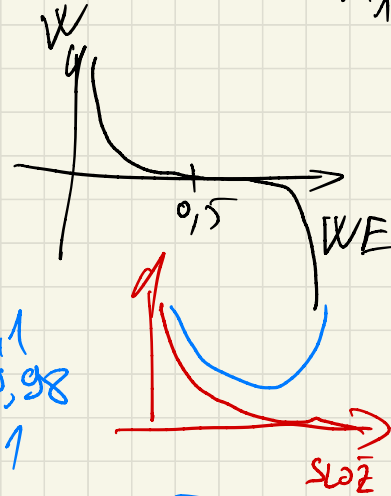
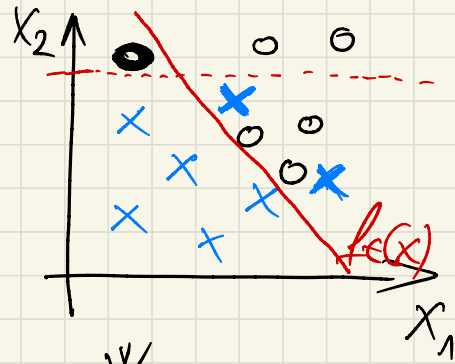
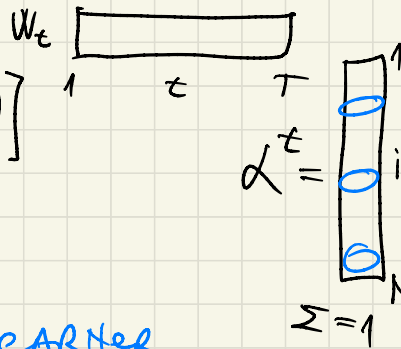
$$\alpha_i = \frac{\alpha_i}{\sum \alpha_i}$$

(NORM)
STABILNOST

$$\hat{y}_t = f_t(x)$$

$$\hat{y} = \text{SIGN}\left[\sum_{t=1}^T W_t f_t(x)\right]$$

$$Y = \{-1, 1\}$$



- $WE = 0.01 \rightarrow W = 2.3$
- $WE = 0.5 \rightarrow W = 0$
- $WE = 0.99 \rightarrow W = -2.3$
- ✓ $W_t = 2.3 \rightarrow \text{FACTOR} = 0.1$
- ✗ $W_t = 2.3 \rightarrow \text{FACTOR} = 9.98$
- ✗ $W_t = 0 \rightarrow \text{FACTOR} = 1$

$$\sum_i \alpha_i^t = \underline{2_t}$$

$$E_{tr}(f) \leq \prod_1^T 2_t$$